

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
4	(a)	(i)		190 + 190 + 290 + 240 (1) [= 910 km]	1			1	1	
		(ii)		Substitution of either: April: $\text{speed} = \frac{\text{distance}}{\text{time}} = \frac{910}{31.5}$ or $\text{speed} = \frac{\text{distance}}{\text{time}} = \frac{610}{19.25}$ (1) April = 28.9 [km/h] (1) June = 31.7 [km/h] (1) [Conclusion - mean speed in June / 31.7 km/h is larger than April / 28.9 km/h so statement correct]			3	3	3	
	(b)	(i)		Substitution: $42 = 8.0 \times 3 + 0.5 \times a \times 3^2$ (1) So: $42 = 24 + 4.5a$ Rearrangement; $a = \frac{42 - 24}{4.5} = \frac{18}{4.5}$ (1) = 4 (1) m/s ² (1)	1 1	1 1		4	3	
		(ii)	I	Substitution: $F = ma$ i.e. $284 = m \times 4$ (ecf) (1) $m = \frac{284}{4} = 71$ [kg] (1)	1	1		2	2	
			II	71 (ecf) – $64.5 = 6.5$ [kg]		1		1	1	
				Question 4 total	4	4	3	11	10	0